

Question Bank – Blockchain Technologies

Unit 1

2-Markers

1. Define Blockchain.
2. Who proposed Blockchain and when?
3. What is the difference between Bitcoin and Blockchain?
4. What is a node in blockchain technology?
5. Define a blockchain wallet.
6. What is blockchain mining?
7. Explain consensus protocol briefly.
8. Differentiate between symmetric and asymmetric encryption.
9. Define hashing in blockchain.
10. What is a Merkle Tree?

5-Markers

1. Explain the importance of cryptography in blockchain with examples.
2. Explain symmetric and asymmetric encryption with suitable examples.
3. Discuss the blockchain transaction process with a suitable example.
4. Explain the Byzantine Generals Problem and its significance in blockchain.
5. What is distributed consensus, and why is it essential in blockchain technology?

Unit 2

2-Markers

1. What makes blockchain an immutable ledger?
2. Define decentralization in blockchain.
3. What is Proof of Burn (PoB)?
4. Briefly explain Proof of Space.
5. What is Proof of Elapsed Time (PoET)?
6. Name different types of data stored on blockchain.
7. What is transaction data in blockchain?
8. Define smart contracts.
9. Explain metadata in blockchain.
10. What is IPFS?

5-Markers

1. Describe how blockchain achieves immutability and why it's significant.
2. Explain the features and benefits of IPFS.
3. Discuss Proof of Burn and provide an example of its application.
4. Explain Proof of Elapsed Time and its operational workflow.
5. Describe hybrid blockchain-database solutions with a real-world example.

Unit 3

2-Markers

1. What is the purpose of blockchain frameworks?
2. Name any two popular blockchain frameworks.
3. Define smart contracts.
4. What is Ethereum Virtual Machine (EVM)?
5. Explain the concept of “Gas” in Ethereum.
6. Define Decentralized Autonomous Organizations (DAOs).
7. Why did Ethereum split into ETH and ETC?
8. Mention two Ethereum components.
9. Define mining nodes in Ethereum.
10. What is Solidity?

5-Markers

1. Explain blockchain frameworks, their components, and their advantages.
2. Discuss Ethereum’s smart contracts with real-world usage examples.
3. Describe the Ethereum DAO incident and its impact.
4. Explain Ethereum architecture with suitable diagrams.
5. Discuss Ethereum Virtual Machine (EVM) and its role in blockchain operations.

Unit 4

2-Markers

1. Define Hyperledger.
2. What is unique about Hyperledger compared to other blockchains?
3. Name two Hyperledger frameworks.
4. What is chaincode in Hyperledger?
5. Explain Hyperledger Composer briefly.
6. Define endorsement policies in Hyperledger.
7. What is permission-based blockchain?
8. Name any two nodes used in Hyperledger.
9. Define the transaction layer in Hyperledger.
10. Explain consensus briefly as used in Hyperledger.

5-Markers

1. Describe Hyperledger architecture with a suitable diagram.
2. Explain the transaction flow in Hyperledger.
3. Discuss the importance and types of endorsement policies in Hyperledger.
4. Discuss consensus mechanisms used in Hyperledger.
5. Discuss various use cases of Hyperledger technology in different industries.

Unit 5

2-Markers

1. What makes Corda unique compared to traditional blockchains?

2. Mention two features of EOSIO.
3. Define parachains in Polkadot.
4. What is XCMP in Polkadot?
5. Define Factom and its core purpose.
6. Explain the concept of cross-blockchain transfers.
7. What are decentralized autonomous corporations (DACs)?
8. Name two key features of Factom.
9. Mention one use-case for EOSIO.
10. Define relay chain in Polkadot.

5-Markers

1. Explain Corda's features and its use cases in financial services.
2. Describe EOSIO's consensus model and its applications.
3. Discuss how blockchain transforms supply chain management.
4. Explain Factom's approach to data security and immutability.
5. Discuss the impact of blockchain frameworks like Polkadot and Factom in healthcare and supply chain management.